
Windsock is Dancing: Adaptive Multimodal Retrieval-Augmented Generation

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Abstract

Multimodal Retrieval-Augmented Generation (MRAG) has emerged as a promising method to generate factual and up-to-date responses of Multimodal Large Language Models (MLLMs) by incorporating non-parametric knowledge from external knowledge bases. However, existing MRAG approaches suffer from static retrieval strategies, inflexible modality selection, and suboptimal utilization of retrieved information, leading to three critical challenges: determining **when** to retrieve, **what** modality to incorporate, and **how** to utilize retrieved information effectively. To address these challenges, we introduce Windsock, a query-dependent module making decisions on retrieval necessity and modality selection, effectively reducing computational overhead and improving response quality. Additionally, we propose Dynamic Noise-Resistance (DANCE) Instruction Tuning, an adaptive training strategy that enhances MLLMs’ ability to utilize retrieved information while maintaining robustness against noise. Moreover, we adopt a self-assessment approach leveraging knowledge within MLLMs to convert question-answering datasets to MRAG training datasets. Extensive experiments demonstrate that our proposed method significantly improves the generation quality by 17.07% while reducing 8.95% retrieval times.

1 Introduction

While Large Language Models (LLMs) have demonstrated remarkable capabilities [1, 2, 3], they often struggle with factual accuracy [4, 5], leading to the development of Retrieval-Augmented Generation (RAG) as a solution by incorporating external knowledge into the generation process [6]. In a typical RAG pipeline, given a query, the system first retrieves relevant documents from an external knowledge base and then concatenates them with the query as context, enabling the model to generate responses grounded in the retrieved information.

While RAG has shown significant success with text-based knowledge [7, 8], real-world tasks inherently require multimodal understanding, like visual analysis or spatial reasoning [9]. This fundamental need for multimodal comprehension has led to the development of multimodal RAG (MRAG) systems that can seamlessly integrate both textual and visual knowledge in their reasoning process [10].

However, existing MRAG methods suffer from three significant limitations. First, they adopt a retrieve-for-all strategy that indiscriminately retrieves information for every query and cannot determine **when** to retrieve, even when the model’s parametric knowledge is sufficient to generate accurate responses [11, 12]. It leads to unnecessary computational overhead and may introduce noise from unreliable retrievers that degrade response quality. Second, current approaches either consistently

retrieve images [11, 12] or exclusively access text-based knowledge bases like Wikipedia [13, 14] without considering the specific informational needs of each query. This inflexible approach fails to recognize that different queries may require different types of information for optimal response generation, underscoring the necessity for an adaptive retrieval mechanism to make query-dependent decisions about **what** modality to select. Third, Multimodal Large Language Models (MLLMs) [15, 16] may struggle to know **how** to utilize retrieved information effectively and are sensitive to irrelevant documents [11, 17]. Either statistic-based [18] or vector-based [19, 20] retrievers may wrongly understand the query and return irrelevant documents, leading to reduce the reliability of responses and the risk of factual errors and hallucinations [21, 22].

To address these limitations, we propose Windsock, a query-aware decision module that dynamically orchestrates the retrieval process for the when and what challenges. Specifically, Windsock decides if a retrieval operation is necessary and what modality should be retrieved based on user queries. For example, queries about historical events are more likely to benefit from textual documents, while visual documents contribute to queries regarding artist’s painting styles. This adaptive approach reduces computational overhead and improves response accuracy by ensuring retrieved information aligns with query intentions. Furthermore, Windsock’s modular design allows it to be seamlessly integrated into open-source and proprietary MLLMs. To address the how challenge, we devise Dynamic Noise-Resistance (DANCE), an adaptive training strategy to dynamically select a modality during instruction tuning in response to retrieval noises, further enhancing MLLMs to improve their utilization of retrieved information while improving their robustness against irrelevant information.

Furthermore, we propose an automatic self-assessment approach to convert any question-answering dataset to support RAG training. Specifically, we assess the inherent ability of MLLMs to identify their incomplete knowledge by providing or withholding retrievals with different modalities. Then, we can evaluate responses to identify the best retrieval strategy and the challenging samples for training.

To summarize our contributions, we propose Windsock, a lightweight module that makes query-dependent decisions on retrieval necessity and modality selection, effectively reducing computational overhead while improving response quality. We develop DANCE, an adaptive training strategy that enhances MLLMs’ ability to utilize retrievals while maintaining robustness against noise. Additionally, we introduce a self-assessment pipeline that automatically converts question-answering datasets for RAG training without relying on proprietary models, enabling efficient identification of optimal retrieval strategies and instruction tuning dataset construction. Comprehensive experiments demonstrate that our proposed method significantly improves both efficiency and response quality.

2 Related Work

Multimodal Retrieval-Augmented Generation Multimodal Retrieval-Augmented Generation (MRAG) has emerged as a promising approach to enhance multimodal large language models (MLLMs) by incorporating external knowledge sources. Early works focused on optimizing retrieval mechanisms through end-to-end training. RA-CM3 [6] and MuRAG [23] pioneered simultaneous optimization of retrievers and generators, while RagVL [12] introduced reranking through MLLM fine-tuning. Subsequently, WikiLLaVA [24] and EchoSight [25] advanced entity-centric retrieval by identifying and leveraging relevant contextual information. SURf [11] introduced selective information utilization to improve knowledge integration and enhance noise rejection. RIR [26] innovated by incorporating real-time information from online search engines. VisRAG [27] and VisDoM [28] specialized in multimodal document retrieval, and Re-ViLM [29] enhanced image captioning by integrating external knowledge into transformer cross-attention layers. Domain-specific applications have demonstrated MRAG’s practical value, with systems like FactMM-RAG [30] improving medical radiology report generation. Recent benchmarks, including MRAG-Bench [17] and SnapNTell [31], have highlighted both the potential and limitations of MRAG systems, particularly in handling entity-centric queries and utilizing retrieved visual knowledge effectively. Most approaches employ a retrieval-for-all strategy, leading to computational overhead and potential noise injection. While recent works like ReflectiVA [14] and mR²AG [13] have introduced adaptive retrieval and reranking through special tokens, they rely on expensive human or GPT-4 annotations and overlook the importance of modality selection based on query requirements. Our approach addresses these limitations by query-dependent decisions for both retrieval necessity and modality selection, thereby optimizing computational efficiency and response quality.

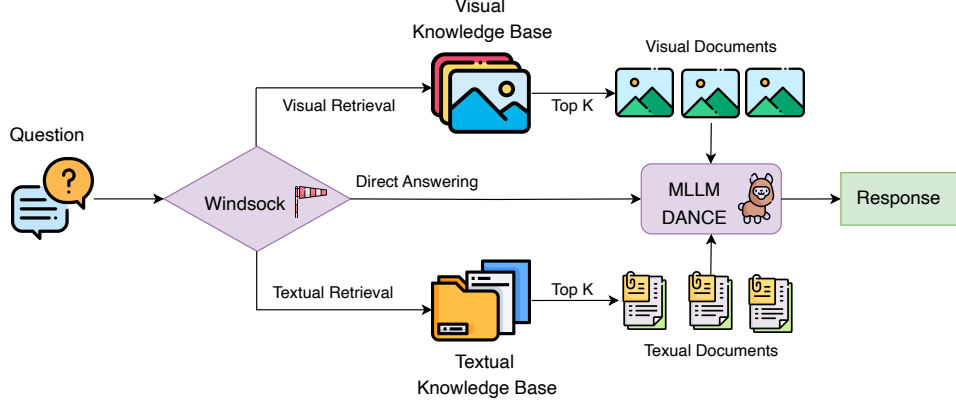


Figure 1: Framework overview of our proposed method. Given a query, Windsock adaptively selects between direct answering (no retrieval) or retrieving from either visual or textual knowledge bases, followed by MLLM instruction tuned by DANCE, generating the final response based on the query and retrieved documents.

Multimodal Large Language Models Recent advances in Multimodal Large Language Models (MLLMs) have enabled systems to process and generate both visual and textual information seamlessly. Leading commercial models such as GPT-4 [15] and Gemini [2] have demonstrated remarkable capabilities across various tasks. LLaVA [16] and PaLM-E [32] have further advanced the field by introducing efficient training methods that effectively leverage pre-trained language models and visual encoders. Particularly noteworthy is the emergence of open-source alternatives like LLaMA-Vision [3], InternVL [33], and QwenVL [34], opening access to multimodal capabilities while achieving competitive performance on standard benchmarks. In this work, we focus on enhancing MLLMs’ capabilities for MRAG tasks.

3 Method

3.1 Task Definition

Given an input query Q consisting of text T and/or image I , Multimodal Retrieval-Augmented Generation aims to generate a response r by leveraging an external knowledge base $\mathbb{D}$, a retriever $\mathcal{R}$, and a multimodal large language model $\mathcal{G}$. The knowledge base $\mathbb{D} = \{d_1, d_2, \dots, d_n\}$ contains a collection of documents. RAG involves two key steps. (1) Retrieval: $\mathcal{R}(Q, \mathbb{D}) \rightarrow \mathbb{D}'$, where $\mathbb{D}' = \{d_1, d_2, \dots, d_k\}$ that identifies the k most relevant documents from $\mathbb{D}$ given the query Q , and (2) Generation: $\mathcal{G}(Q, \mathbb{D}') \rightarrow r$ that produces the final response r by conditioning on both the original query Q and the retrieved documents. The objective is to maximize the quality and factual accuracy of the generated response r while maintaining coherence with both the input query and the retrieved knowledge. Unlike previous works employing a static retrieval strategy [24, 25, 11], our work groups each modality into different knowledge bases and use a query-dependent module to trigger the retrieval and select the modality dynamically. To simplify but without loss of generality, we consider visual knowledge base $\mathbb{D}^I$ containing images (and/or their metadata) and textual knowledge base $\mathbb{D}^T$ including text documents.

3.2 Overview

As illustrated in Figure 1, given a query Q , the Windsock module decides if the retrieval function $\mathcal{R}(Q, \mathbb{D})$ is called. If Q can be answered without external knowledge, we directly use $\mathcal{G}$ to generate a response to reduce the retrieval latency and context length. Otherwise, Windsock decides the retrieved modality type based on the query, and the retriever returns the documents from the knowledge base of the corresponding modality. Then, a response is generated based on the query and retrieved information. Moreover, we use Dynamic Noise-Resistance (DANCE) Instruction Tuning to enhance the ability to utilize retrieved documents.

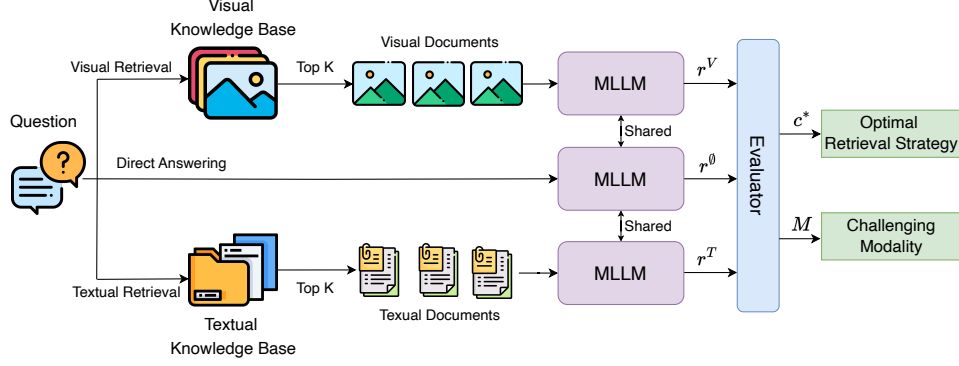


Figure 2: Data construction pipeline. An MLLM generates responses using different strategies (direct answering, visual retrieval, and textual retrieval), which are then evaluated to determine both the optimal retrieval strategy and challenging modality for training.

3.3 Windsock as an Adaptive Information Retriever

To address the limitations of static retrieval strategies in multimodal RAG systems, we propose Windsock, an adaptive retrieval mechanism that optimizes both retrieval necessity and modality selection. Unlike conventional approaches that indiscriminately retrieve information for every query [11], Windsock determines when external knowledge is required and what type of information would be most beneficial for response generation. Specifically, Windsock implements a three-way classification that maps input queries to optimal retrieval strategies. Given a user query Q , Windsock $\mathcal{W}$ generates a retrieval type c through the mapping:

$$c = \mathcal{W}(Q) \in \{\text{NA}, \text{Visual}, \text{Textual}\}. \quad (1)$$

Based on the retrieval type, a response is generated by:

$$\begin{cases} r^\emptyset = \mathcal{G}(Q, \emptyset), & \text{if } c = \text{NA}, \\ r^V = \mathcal{G}(Q, \mathcal{R}(Q, \mathbb{D}^I)), & \text{if } c = \text{Visual}, \\ r^T = \mathcal{G}(Q, \mathcal{R}(Q, \mathbb{D}^T)), & \text{if } c = \text{Textual}. \end{cases} \quad (2)$$

When the retrieval type is “NA,” the query can be adequately answered using the MLLM’s parametric knowledge, and Windsock bypasses retrieval entirely; when the retrieval type is “Visual,” the query requires visual context and Windsock activates visual document retrieval to obtain the context; when the retrieval type is “Textual,” Windsock triggers text-based retrieval to fetch text documents. The hybrid retrieval and other new modalities can be supported by adding new retrieval types in Equation (2). More discussions can be found in Section G.

Discussion This adaptive approach offers several advantages over traditional static retrieval methods. First, it reduces computational overhead by avoiding unnecessary retrievals when the MLLM’s parametric knowledge suffices. Second, bypassing retrieval for queries that can be answered directly prevents potential degradation of response quality due to noisy or irrelevant retrievals [35]. Third, while transformers process both visual and textual inputs as tokens, different types of information are inherently better represented in their native modalities due to the way knowledge is structured in datasets. For example, spatial relationships might be more efficiently captured in visual form, while historical facts may be better preserved in text. Our analysis reveals that downstream datasets exhibit such implicit biases in their information requirements, which Windsock effectively captures to optimize retrieval strategy.

3.4 Training Dataset Curation

A critical challenge is obtaining high-quality training data that indicates the optimal retrieval strategy for each query to train the Windsock. Rather than relying on expensive annotations from commercial models [15, 2], we propose a self-assessment approach that leverages the inherent capabilities of MLLMs to automatically determine the most effective retrieval strategy, as shown in Figure 2.

For each question-answer pair $\{Q, A\}$ in a given question-answering dataset, where Q and A are question and answer, respectively, we generate three distinct responses by employing different retrieval strategies, following Equation (2). To evaluate the effectiveness of each strategy, we compute quality scores for these responses by comparing them against the ground truth answer:

$$\begin{aligned} s^{\mathcal{O}} &= \epsilon(r^{\mathcal{O}}, A) \\ s^I &= \epsilon(r^I, A) \\ s^T &= \epsilon(r^T, A), \end{aligned} \tag{3}$$

where ϵ represents an evaluation function specific to the downstream task requirements, e.g., F1 score.

Based on these scores, we determine the optimal retrieval strategy c^* for each query:

$$c^* = \arg \max_c (s^{\mathcal{O}}, s^I, s^T). \tag{4}$$

The training set of Windsock is constructed as $\{Q, c^*\}$.

Discussion The key advantage of this self-assessment method is that it directly optimizes task performance while accounting for the specific strengths and limitations of the underlying MLLM. By comparing performance across different retrieval strategies, we can identify cases where retrieval might be detrimental (when $s^{\mathcal{O}} > \max(s^I, s^T)$) or where specific modalities are particularly beneficial (when either s^I or s^T outperforms others).

3.5 Dynamic Noise-Resistance Instruction Tuning

While Windsock effectively determines when and what information to retrieve, MLLMs still face challenges in utilizing retrieved information effectively, particularly when dealing with irrelevant content from unreliable retrievers or inappropriate chunk sizes. Traditional instruction tuning approaches are inadequate for preparing models to handle the varying quality of retrieved information [36]. To address this limitation, we propose Dynamic Noise-Resistance (DANCE) Instruction Tuning, an adaptive training strategy that enhances MLLMs’ ability to ignore misleading information.

DANCE evaluates responses with different types of retrieved knowledge (visual vs. textual) and dynamically selects the modality where the MLLM struggles most:

$$\arg \min_M (s^I, s^T) \in \{I, T\}, \tag{5}$$

where M denotes the retrieval modality of the challenging modality. The documents corresponding to the selected modality are probably irrelevant. Note that if s^I is equal to s^T , we randomly select one as the challenging modality. Since our primary goal is to improve the robustness of MRAG, we specifically focus on utilizing s^I and s^T and ignore the $s^{\mathcal{O}}$ in the training dataset construction process. These identified challenging cases are then collected to create an instruction tuning dataset in the format $\{Q, \mathcal{R}(Q, \mathbb{D}^M), A\}$. Subsequently, we employ the standard instruction tuning pipeline to fine-tune the MLLMs, following the approach established in previous work [16].

Discussion The injection of noise during instruction tuning has been shown to improve model performance [37]. While recent works have proposed noise injection for robust RAG [12, 22, 36], they rely on ground truth documents provided by dataset annotators or proprietary LLM-as-a-Judge [38]. In contrast, our DANCE approach enables noise-robust training by leveraging self-assessment modality selection. Unlike SURf [11], which uses image similarity for hard example mining and requires sequential document-by-document response generation, our method efficiently identifies challenging cases through downstream tasks’ evaluation metrics and processes different modalities in parallel, resulting in faster and more task-oriented training, as shown in the following section.

4 Experiments

4.1 Experimental Setup

Datasets and Evaluation Metrics We evaluate our proposed method on WebQA [39] and Multi-modalQA [40]. WebQA is a benchmark dataset for multi-modal and open-domain question answering.

Table 1: Main Results on WebQA and MultimodalQA. *Single* and *Multiple* denote one-hop and multi-hop questions in WebQA. [†] denotes a different setting in RagVL paper that only uses validation set images to construct the visual knowledge base. For baselines without Windsock, we provide visual and textual retrievals and report the higher performance.

Method	Generator	WebQA			MultimodalQA	
		Single	Multiple	All	F1	EM
<i>w.o. Instruction Tuning</i>						
Zero-Shot	LLaVA-v1.5-7B	59.79	35.20	42.12	29.94	27.20
	LLaVA-v1.5-13B	58.75	38.14	43.95	34.71	30.93
	Qwen2-VL-7B	61.76	37.09	44.04	29.37	26.38
Vanilla RAG	LLaVA-v1.5-7B	58.53	34.36	41.17	35.88	32.83
	LLaVA-v1.5-13B	59.85	35.92	42.66	44.23	40.16
	Qwen2-VL-7B	62.96	38.36	45.29	34.37	31.44
Windsock (Ours)	Qwen2-VL-7B	65.92	38.63	46.32	46.90	43.01
<i>w. Instruction Tuning</i>						
RagVL [†] [12]	LLaVA-v1.5-13B-RagVL	57.06	76.18	65.56	-	76.96
DANCE [†] (Ours)	LLaVA-v1.5-13B-DANCE	67.02	73.70	69.98	-	77.39
SURf [11]	Qwen2-VL-7B-SURf	62.72	55.60	57.61	50.98	46.23
DANCE (Ours)	Qwen2-VL-7B-DANCE	66.42	57.45	59.97	51.62	47.01
Windsock+DANCE (Ours)	Qwen2-VL-7B-DANCE	70.12	59.32	62.36	52.72	48.20

Table 2: Effectiveness and efficiency of Windsock. Performance comparison using different retrieval strategies measured by average inference time per query and F1 score on WebQA.

Retrieval	Time (s) ↓	Single ↑	Multiple ↑	All ↑
<i>w.o. DANCE</i>				
NA	0.46	61.76	37.09	44.04
Visual	0.67	<u>64.88</u>	36.46	44.47
Textual	0.79	52.87	36.70	41.25
Windsock	<u>0.56</u>	65.92	38.63	46.32
<i>w. DANCE</i>				
NA	0.29	65.58	47.22	52.39
Visual	0.50	63.10	46.58	51.23
Textual	<u>0.33</u>	66.26	<u>58.77</u>	<u>60.88</u>
Windsock	0.36	70.12	59.32	62.36

It comprises over 34,000 training question-answer pairs, covering diverse domains and requiring systems to reason over text snippets and images to generate answers. MultimodalQA is a challenging question-answering dataset that requires joint reasoning with text, tables, and images. It consists of 29,918 questions designed to simulate complex scenarios where information is integrated from multiple sources. As mentioned in the task definition, we construct knowledge bases using images and texts. Our framework can also be extended to the tabular knowledge base. For evaluation, we employ F1 score for WebQA, and both F1 score and Exact Match (EM) for MultimodalQA. These metrics assess the overlap between generated responses and ground truth answers by comparing key entities.

Baseline Methods We compare our approach to SOTA MRAG methods, including RagVL [12] and SURf [11]. Note that RagVL constructs the training set using ground truth annotated documents, which differ from our method. Additionally, instead of constructing an entire knowledge base (403,277 in WebQA and 57,052 in MultimodalQA), RagVL constructs a separate small knowledge base (4,972 in WebQA and 384 in MultimodalQA) for validation, significantly reducing the challenging in retrieval. To make a fair comparison, we replace the original MLLM with our fine-tuned MLLM while keeping all other experimental settings unchanged. Discussions about ReflectiVA, enc-vqa [41] dataset, and infoseek [42] dataset can be found in Section I.

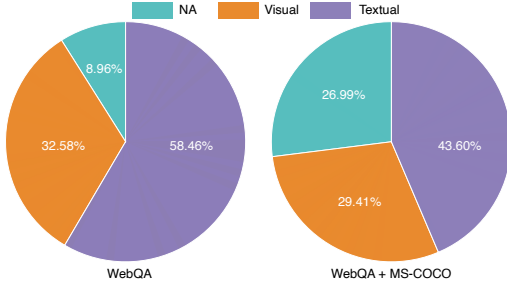


Figure 3: Windsock ratio of retrieval decisions on WebQA and WebQA+MS-COCO, showing the model’s adaptive behavior across different complexities in downstream tasks.

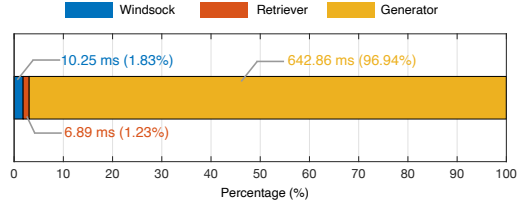


Figure 4: Pipeline runtime breakdown of Qwen2-VL, showing the proportion of total inference time consumed by each component.

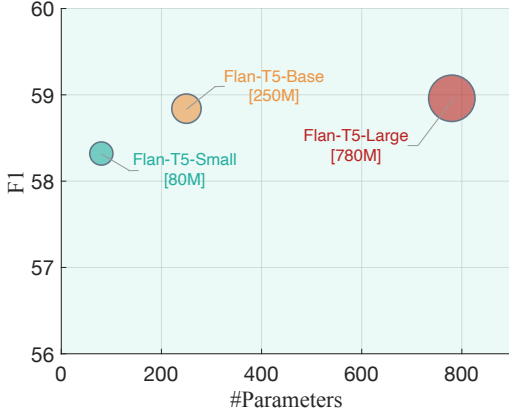


Figure 5: Performance scaling with backbone model size on WebQA, comparing F1 scores versus the number of parameters.

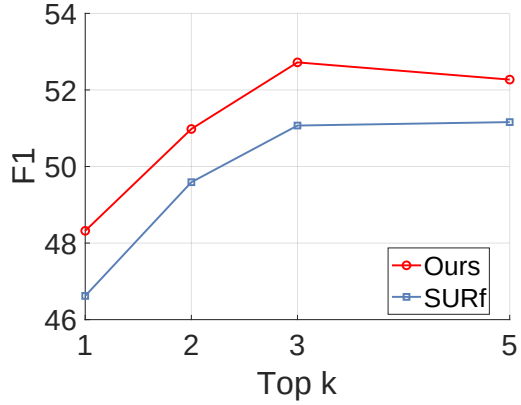


Figure 6: Impact of different top-k values on MultimodalQA.

Implementation Details We use LLaVA [16] and Qwen2-VL [34] series MLLMs as the generator. The retriever is VBGE-base [19] and returns the top 3 retrievals. We use Flan-T5-Small [43] as the backbone of Windsock. To train the Windsock, we use the AdamW [44] optimizer with the learning rate $5e-4$. The batch size is 16, and the training epoch is 5. We use a linear learning rate scheduler, in which the learning rate is down to 0 at the end of training. We compute the class weights and use them in the cross-entropy loss to balance the training data [45]. For DANCE instruction tuning, we use the default LoRA [46] settings in LLaMA-Factory [47], except for setting the training epoch to 1. If not specified, experiments are conducted on WebQA. All training experiments are conducted on 4 NVIDIA H100 GPUs, and inference experiments use 1 NVIDIA H100 GPU.

4.2 Main Results

As shown in Table 1, our method achieves substantial improvements over existing baseline approaches, demonstrating the effectiveness of our proposed method. We observe that Windsock alone provides significant performance gains compared to both zero-shot and vanilla RAG baselines, indicating that our adaptive retrieval strategy effectively reduces noise from unnecessary retrieval while maintaining high response quality. While the performance of different MLLMs varies in different downstream tasks, after instruction tuning, the overall trend in performance is upward. Most notably, combining Windsock with DANCE instruction tuning yields the best performance across all metrics. It improves 16.04% and 5.19% over the *w.o. Instruction Tuning* baseline on WebQA and MultimodalQA, respectively. Moreover, compared to SURf, it shows an improvement of 4.75% and 1.97% on WebQA and MultimodalQA, demonstrating the effectiveness of our DANCE instruction tuning method. Our proposed method can even outperform RagVL trained with ground truth documents in several metrics, showing great potential in training with noise for robust RAG systems.

Table 3: Ablation study of Windsock variants on WebQA, showing the impact of modality selection and classification strategy.

Variant	Single	Multiple	All
Windsock	65.92	38.63	46.32
w.o. Modality Selection	62.68	36.76	43.89
w.o. One-stage	61.31	37.39	44.13

Table 5: Comparison of different instruction tuning strategies on WebQA.

Strategies	Single	Multiple	ALL
Easy	58.83	51.24	53.38
Random	60.98	53.94	55.12
Ours	66.42	57.45	59.97

Table 4: Performance comparison of different MLLMs as generators on WebQA with Windsock.

Methods	Single	Multiple	Overall
LLaVA-v1.5-7B	59.83	37.10	43.75
LLaVA-v1.5-13B	59.91	38.61	44.48
Qwen2-VL-7B	65.92	38.63	46.32

Table 6: Performance comparison on the MME benchmark.

MLLM	Perception	Cognition
Qwen2-VL-7B	1423.25	484.00
Qwen2-VL-7B-DANCE	1394.05	435.71
LLaVA-v1.5-13B	1469.46	294.64
LLaVA-v1.5-13B-DANCE	1377.36	309.64

4.3 Analysis on Windsock

Effectiveness of Windsock Table 2 shows the results with different retrieval types. We notice that only using a single modality knowledge base cannot consistently improve performance, and different queries may require different modality information. Moreover, our proposed Windsock outperforms them by dynamically deciding when to retrieve and what modality is useful based on the query.

Efficiency of Windsock We present the average inference time per question in Table 2. While *NA* achieves the fastest inference time by omitting document retrieval, its performance is unsatisfactory. Notably, *w. DANCE* demonstrates lower inference times compared to *w.o. DANCE*. This efficiency gain stems from the shorter answer lengths in downstream tasks. WebQA typically requires phrase-level or single-sentence responses, which are more concise than the text lengths used during pre-training [3] or supervised fine-tuning stage [16]. Through instruction tuning, MLLMs learn to generate responses that align with the downstream dataset’s answer length characteristics [48], typically shorter than the original MLLM outputs. Since MLLMs generate text in an autoregressive way, shorter responses require fewer token generation steps, resulting in reduced inference time. Detailed response length analysis can be found in the Appendix. Our proposed Windsock effectively balances performance and efficiency through dynamic computation allocation.

Decisions from Windsock To analyze Windsock’s decision-making patterns, we visualize the distribution of retrieval decisions in Figure 3. We evaluate Windsock on the WebQA validation set and an additional 2,000 question-answering pairs sampled from the MS-COCO validation set [49]. MS-COCO samples represent simpler queries that can be answered without external knowledge, contrasting to WebQA’s knowledge-intensive questions. Our analysis reveals that Windsock skips retrieval for 8.96% of WebQA queries. When including simpler MS-COCO queries, the skip rate increases to 26.99%, demonstrating that Windsock effectively adapts its retrieval strategy based on query complexity and dataset characteristics.

Overheads of Windsock Figure 4 illustrates the computational breakdown of each stage in the MRAG pipeline. The generator dominates the computational cost, accounting for 642.86 ms (96.94%) of the total inference time. Windsock only adds 10.25 ms (1.83%) overhead and it effectively reduces the overall computational burden by selectively skipping unnecessary retrievals, significantly improving inference speed, as shown in Table 2.

Scalability of Windsock Figure 5 demonstrates how Windsock’s performance scales with backbone model size, showing consistent performance improvements as the number of parameters increases. More experiments can be found in Section F. We use the language model, Flan-T5, as Windsock’s backbone based on prior work [50], which shows that textual queries are crucial for understanding

Table 7: Performance comparison using ground truth documents on WebQA validation set.

Method	F1
Qwen2-VL-7B	50.10
Qwen2-VL-7B-DANCE	67.35
LLaVA-v1.5-13B	44.68
LLaVA-v1.5-13B-DANCE	64.07

Table 8: Pipeline efficiency comparison measuring dataset construction time (in GPU hours).

Method	GPU Hours ↓
SURf	32.45
Ours	15.43

user intent and question complexity. Nevertheless, the Windsock framework can be extended to multimodal backbones in future work.

Variants of Windsock To further verify the effectiveness of Windsock, we compare it to several variants, as illustrated in Table 3. *w.o. Modality Selection* denotes removing the dynamic modality selection in Windsock and keeping the ability to decide when to retrieve. The performance drop demonstrates the importance of modality-specific retrieval. *w.o. One-stage* represents a two-step decision process in which we train two separate classifiers. The first is used to decide when to retrieve, and the second is employed to determine the retrieved modality. Its lower performance suggests that joint optimization in a single stage is more effective.

More discussions about the Windsock and self-assessment framework can be found in Section H.

4.4 Analysis on DANCE

Backbones of Generators Table 4 shows the results of different MLLMs as the generator. The results represent that stronger MLLMs could also achieve better performance on the MRAG task. We believe that with more SOTA MLLMs released, our proposed method could also be improved.

Analysis of Instruction Tuning Strategy To evaluate the effectiveness of our DANCE instruction tuning approach, we compare it with several strategies. As shown in Table 5, we consider three alternative approaches. *Easy* denotes we select the modality M with the higher score in Equation (5). *Random* means randomly selecting a modality for tuning. Our proposed method achieves the best performance across all metrics, demonstrating that dynamically selecting challenging modalities leads to more effective instruction tuning.

Impact of Retrieval Size To investigate the impact of retrieval size on model performance, we conducted experiments varying the number of retrieved documents. As shown in Figure 6, both our method and SURf show improved performance as k increases, with diminishing returns after $k = 3$. Our approach achieves optimal F1 scores at $k = 3$.

Performance on General MLLM Benchmark To verify how DANCE instruction tuning affects the general abilities of MLLMs, we report the results on MME benchmark [51], as illustrated in Table 6. The results show a trade-off between the abilities of downstream tasks and MLLMs’ general abilities. We also notice that after DANCE tuning, the cognition score of LLaVA-v1.5-13B-DANCE outperforms the original MLLM, showing there is a potential to improve the cognition ability of LLaVA models. Discussions about improving the general ability can be found in Section J.

Performance on Ground Truth Data To isolate the effectiveness of our instruction tuning method in information utilization (independent of retrieval quality), we conducted experiments where ground truth documents are directly provided to MLLMs, as shown in Table 7.

Efficiency of Data Creation Pipeline Table 8 compares dataset construction time between our method and SURf, as both approaches convert question-answering datasets to RAG format without requiring explicit RAG annotations. While SURf processes documents sequentially, generating one response per document, our pipeline processes k documents per modality simultaneously. This parallel processing approach significantly reduces dataset construction time.

More discussions about the ability of noise rejection can be found in Section B.

5 Conclusion

In this paper, we presented Windsock, an adaptive approach to multimodal retrieval-augmented generation that dynamically determines retrieval necessity and modality selection. Combined with our DANCE instruction tuning method, this approach significantly improves both computational efficiency and response quality. Through query-dependent retrieval decisions and enhanced utilization of retrieved information, our method advances the state-of-the-art in MRAG systems while reducing computational overhead.

6 Acknowledgments

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